

PROJECT REPORT CSA2001

AI-BASED WATER QUALITY ANOMALY DETECTION SYSTEM

Using Unsupervised Machine Learning

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Taranija Singh

Abstract

Water-quality monitoring produces multiple physical and chemical measurements at the same time. A single parameter may appear ordinary while the combination of several parameters can form an unusual pattern. This project develops an unsupervised machine-learning system for detecting such statistically unusual water-quality observations. The system uses the Kaggle Water Potability dataset containing 3,276 observations and nine numerical water-quality features. Missing values are handled using median imputation and the resulting features are standardized before modelling. Isolation Forest is used as the primary anomaly detector, with Local Outlier Factor (LOF) and One-Class Support Vector Machine (One-Class SVM) included for comparison. The trained models are serialized with Joblib and exposed through both a command-line interface and an interactive Streamlit web application. The primary Isolation Forest model flags 164 of 3,276 observations as anomalous at a contamination setting of 5%, while LOF flags 132 and One-Class SVM flags 170. The project also produces anomaly-score, feature-distribution, model-comparison, and PCA visualization plots.

Keywords

Water quality, anomaly detection, unsupervised learning, Isolation Forest, LOF, One-Class SVM, Streamlit, Python, machine learning.

1. Introduction

Water quality is influenced by several interacting chemical and physical properties such as pH, dissolved solids, hardness, sulfate concentration, conductivity, organic carbon, trihalomethanes and turbidity. Monitoring these variables manually across many observations can make unusual combinations difficult to identify. Anomaly detection provides a way to learn the structure of observed data and highlight samples that are statistically different from the dominant pattern.

This project focuses on anomaly detection rather than directly predicting potability. The system therefore does not claim that an anomalous sample is automatically unsafe, contaminated, or non-potable. Instead, it identifies an

observation that is unusual relative to the patterns learned from the training dataset.

2. Problem Statement

Develop an automated machine-learning system that can analyze multivariate water-quality measurements and identify observations that deviate substantially from the learned data distribution without depending on anomaly labels during training.

3. Motivation

- Water quality depends on interactions among multiple variables rather than a single measurement.
- Real-world anomaly labels can be scarce, making unsupervised learning useful for exploratory monitoring.
- Statistical outliers may indicate unusual environmental conditions, sensor behavior, or process changes that deserve investigation.
- A reproducible machine-learning pipeline makes the analysis repeatable and easier to demonstrate.
- An interactive interface makes the trained models accessible without requiring the evaluator to write Python code.

4. Objectives

1. Prepare a reproducible preprocessing pipeline for numerical water-quality observations.
2. Handle missing values without manually fabricating measurements.
3. Train Isolation Forest as the primary unsupervised anomaly detector.
4. Compare Isolation Forest with LOF and One-Class SVM.
5. Generate data-driven visualizations of anomaly scores and feature behavior.
6. Save the trained models and preprocessing pipeline for later inference.
7. Provide a command-line application and a Streamlit web interface.
8. Demonstrate normal and anomalous observations using authentic dataset-based examples.

5. Dataset

The project uses the Water Potability dataset stored locally as `data/water_potability.csv`. The dataset contains 3,276 observations, 9 numerical water-quality features, and an optional Potability label. The Potability column is not used as an input to the anomaly-detection models, so the primary modelling task remains unsupervised.

Feature	Meaning	Unit	Missing
ph	Acidity / alkalinity level	pH units	491
Hardness	Calcium and magnesium concentration	mg/L	0
Solids	Total dissolved solids (TDS)	ppm	0
Chloramines	Chloramine concentration	ppm	0
Sulfate	Dissolved sulfate	mg/L	781
Conductivity	Electrical conductivity	$\mu\text{S}/\text{cm}$	0
Organic_carbon	Organic carbon level	ppm	0
Trihalomethanes	Chlorination byproducts	$\mu\text{g}/\text{L}$	162
Turbidity	Water cloudiness	NTU	0

Dataset source documented by the project: Kaggle Water Potability Dataset / public mirror. The repository README identifies the public mirror as https://raw.githubusercontent.com/fenago/datasets/main/water_potability.csv.

6. Data Preprocessing

The preprocessing pipeline is designed so that incoming observations receive the same transformation used during model training. Missing numerical values are filled with the median of the corresponding feature, and `StandardScaler` is applied to standardize the feature matrix. The fitted preprocessing object is saved as `preprocessing_pipeline.pkl`.

- Select the nine numerical water-quality variables.
- Convert numerical fields to numeric form and handle missing values.
- Apply median imputation using `SimpleImputer(strategy='median')`.
- Standardize features using `StandardScaler`.
- Persist the preprocessing pipeline with `Joblib`.
- Apply the same pipeline during prediction.

7. Machine Learning Methodology

The project follows an unsupervised anomaly-detection workflow. The main model is Isolation Forest, supported by two comparison approaches: Local Outlier Factor and One-Class SVM.

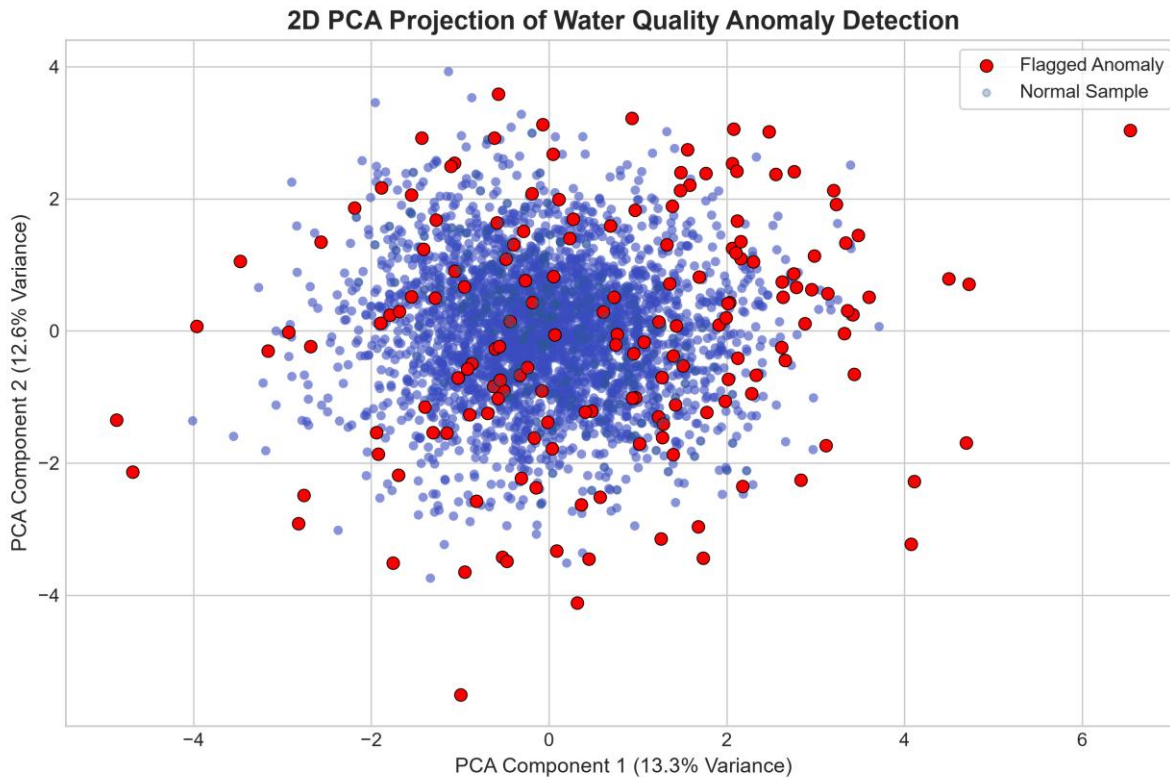


Figure 1. PCA projection generated by the project for visualization of normal and flagged observations.

7.1 Isolation Forest

Isolation Forest builds randomized decision trees that recursively partition the feature space. Observations that are unusual and relatively isolated tend to require fewer partitions to separate from the rest of the data. These shorter path lengths contribute to lower anomaly-related decision scores. In this project, Isolation Forest is the primary model and uses a documented contamination setting of 0.05.

7.2 Local Outlier Factor

LOF compares the local density around an observation with the density of its nearest neighbors. A point whose local density is substantially lower than the surrounding neighborhood can be treated as a local outlier.

7.3 One-Class SVM

One-Class SVM learns a boundary around the dominant region of the feature space. Observations outside that learned region can be classified as outliers. The project uses it as a comparison model rather than the primary detector.

8. System Architecture

The complete workflow is:

Raw water-quality dataset



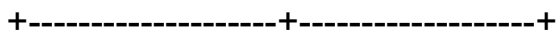
Feature selection (9 numerical parameters)



Median imputation



StandardScaler



Isolation Forest



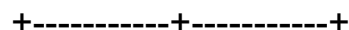
LOF



One-Class SVM



NORMAL / ANOMALOUS result



v
CLI app

v
Streamlit web app

9. Implementation and Project Structure

The repository is organized into data, model, results, source-code and application components.

```
AI-Water-Quality-Anomaly-Detection/  
|  
+-- data/  
|   +-- water_potability.csv  
|  
+-- models/  
|   +-- isolation_forest.pkl  
|   +-- lof_model.pkl  
|   +-- ocsvm_model.pkl  
|   +-- preprocessing_pipeline.pkl  
|  
+-- results/  
|   +-- anomaly_score_distribution.png  
|   +-- feature_distribution.png  
|   +-- model_comparison.png  
|   +-- pca_visualization.png  
|  
+-- src/  
|   +-- preprocess.py  
|   +-- train.py  
|   +-- predict.py  
|   +-- utils.py  
|  
+-- app.py  
+-- main.py  
+-- requirements.txt  
+-- README.md  
+-- Dockerfile
```

+-- .gitignore

9.1 Important Files

File	Purpose
src/preprocess.py	Data cleaning, imputation, scaling and preprocessing pipeline.
src/train.py	Model training, anomaly statistics and visualization generation.
src/predict.py	Loads saved models and performs inference on a water-quality observation.
src/utlis.py	Helper functions, including dataset-based example retrieval.
main.py	Interactive command-line interface.
app.py	Streamlit-based web interface.
models/*.pkl	Serialized trained models and preprocessing pipeline.
results/*.png	Generated visualizations used for analysis and presentation.

10. Evaluation and Results

Because the primary task is unsupervised anomaly detection, conventional supervised accuracy is not used as the main performance measure. Instead, the project reports the number and percentage of observations flagged, score statistics, and agreement among different anomaly detectors.

Model	Anomalies	Percentage	Minimum Score	Maximum Score
Isolation Forest	164	5.01%	-0.1033	+0.1441
Local Outlier Factor	132	4.03%	-0.6387	+0.1872
One-Class SVM	170	5.19%	-0.1983	+9.2415

The primary Isolation Forest model flags 164 observations out of 3,276. This is close to the configured 5% contamination setting. LOF and One-Class SVM provide different anomaly counts because the algorithms use different notions of isolation, local density and decision boundaries.

Comparison	Agreement
Isolation Forest + LOF	90 samples (54.9% of IF anomalies)
Isolation Forest + One-Class SVM	111 samples (67.7% of IF anomalies)
All three models	78 samples

10.1 Generated Visualizations

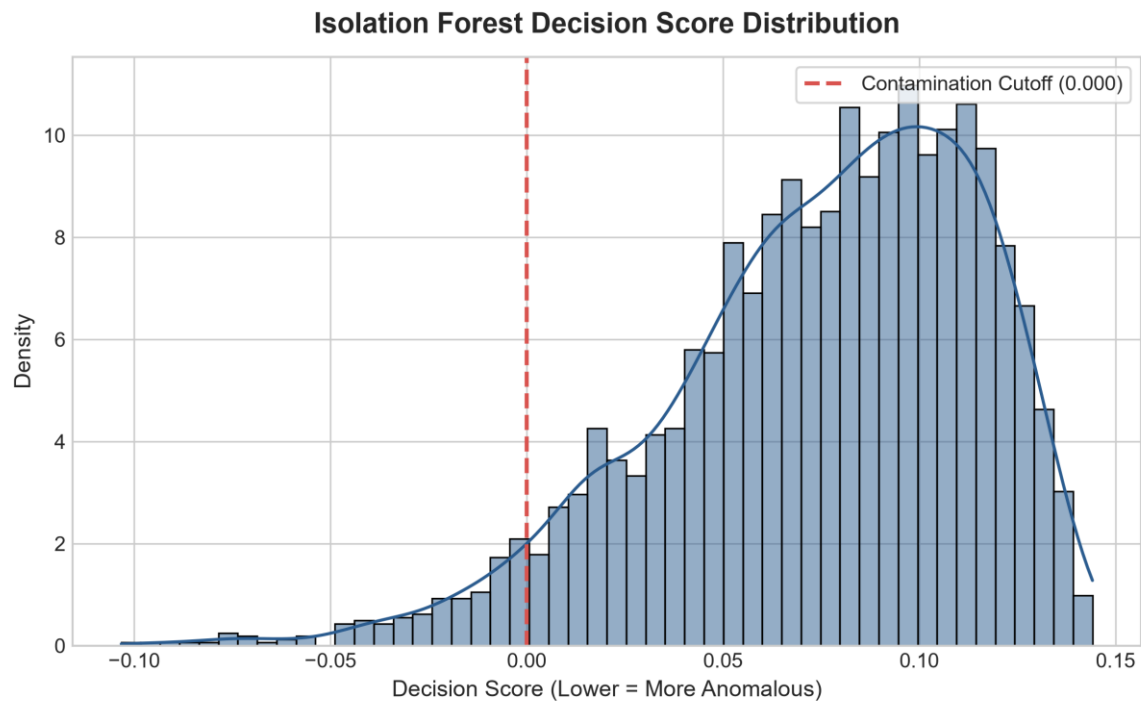


Figure 2. Isolation Forest anomaly-score distribution.

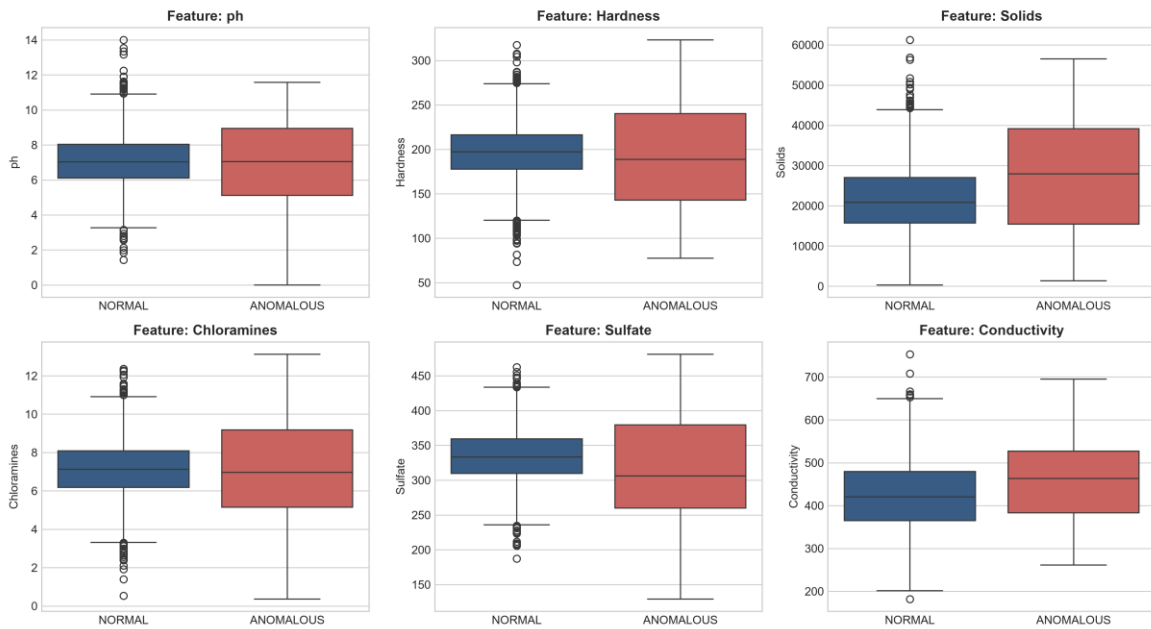


Figure 3. Feature distributions comparing flagged and normal observations.

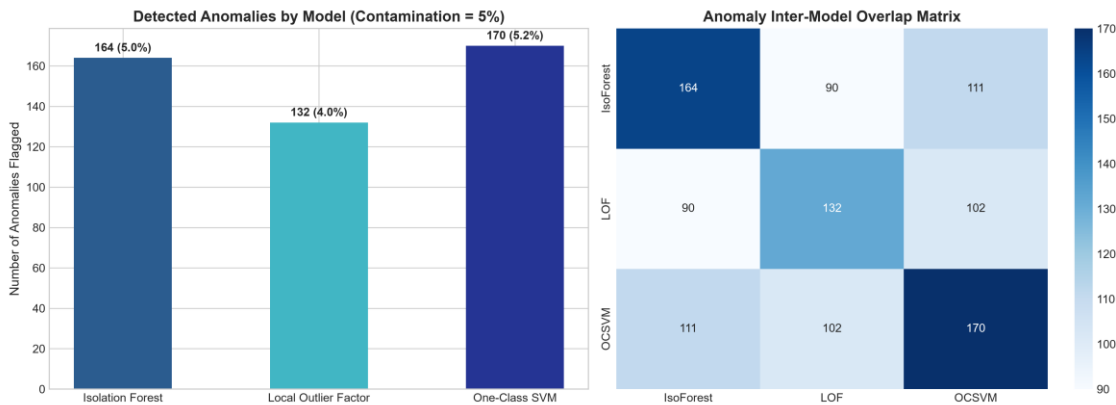


Figure 4. Model anomaly-count and agreement comparison.

11. Demonstration of an Anomalous Observation

The project includes an authentic dataset-based anomalous example so that the demonstration does not depend on inventing arbitrary measurements. When the preset anomalous observation is analyzed, the saved models produce the following documented results.

Feature	Example Value	Interpretation in the Project
pH	8.942	Within the dataset's observed

		range; not the main reason for the anomaly.
Hardness	215.67 mg/L	Within the observed dataset range.
Solids (TDS)	56,488.67 ppm	Extremely high relative to the central dataset distribution.
Chloramines	3.23 ppm	Within the observed dataset range.
Sulfate	129 mg/L	At the low end of the observed dataset range.
Conductivity	541.92 μ S/cm	Within the observed dataset range.
Organic carbon	9.31 ppm	Within the observed dataset range.
Trihalomethanes	70.83 μ g/L	Within the observed dataset range.
Turbidity	4.35 NTU	Within the observed dataset range.

For this preset observation, the project documentation records Isolation Forest as ANOMALOUS with a decision score of -0.1033, LOF as ANOMALOUS, and One-Class SVM as ANOMALOUS. The feature-level analysis highlights the unusually high TDS value and the low-end sulfate value. The model result means that the combined measurement pattern is unusual relative to the training data; it is not a laboratory safety certification.

12. User Interface and Deployment

The application is exposed through Streamlit. Users can choose manual parameter controls or preset dataset examples, review the nine measurements, run the anomaly analysis, and inspect the model diagnosis and pre-computed visualizations.

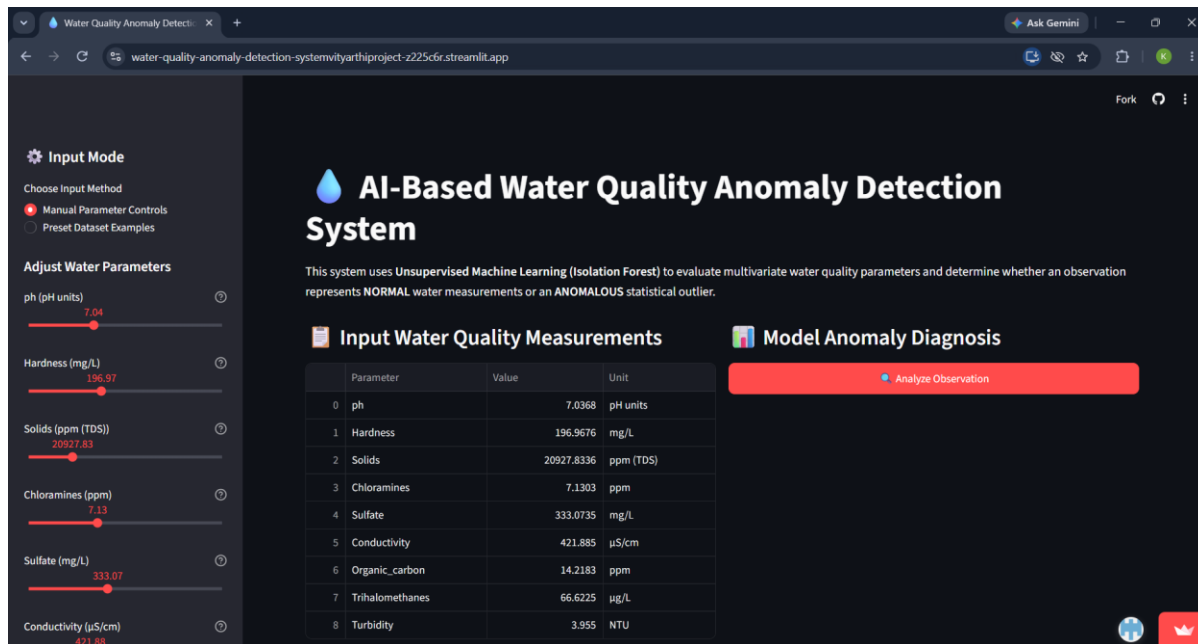


Figure 5. Deployed Streamlit interface opened from an independent browser account.

Live deployment URL: <https://water-quality-anomaly-detection-systemvityarthiproject-z225c6r.streamlit.app/>

The repository used for source-code submission is public on GitHub:
https://github.com/taranijasingh2006-art/Water-Quality-Anomaly-Detection-System_VItyarthy_Project

13. Installation and Execution

The project can be run locally from the repository root.

```
python -m venv venv
venv\Scripts\activate
pip install -r requirements.txt
python src/train.py
python main.py
```

13.1 Streamlit Web Application

```
streamlit run app.py
```

The Streamlit application can then be opened at the local URL shown in the terminal, normally `http://localhost:8501`.

14. Testing Performed

- Verified that the project can load the saved preprocessing pipeline and trained anomaly models.
- Verified the interactive Streamlit interface renders the water-quality controls and model diagnosis area.
- Verified that the application can be opened from a separate browser account, confirming that the deployment is publicly reachable.
- Verified the project contains authentic dataset-based normal and anomalous examples.
- Checked that the three anomaly detectors can provide comparative results for the preset anomalous observation.
- Verified that model outputs are described as anomaly/decision scores rather than probabilities.

15. Limitations

- An anomaly is a statistical deviation from the training data and is not automatically evidence of contamination or unsafe drinking water.
- The dataset is a general water-potability dataset and may not represent every geographic region, season or monitoring station.
- Unsupervised models can identify unusual observations without knowing the real-world cause of the anomaly.
- The contamination parameter influences how many observations are expected to be flagged.
- PCA is a two-dimensional visualization technique and does not replace the original nine-dimensional anomaly model.
- A deployed academic demonstration should not be used as a substitute for laboratory testing or regulatory water-quality assessment.

16. Future Scope

- Integrate live sensor streams for continuous anomaly monitoring.
- Add time-series models to detect sudden changes and persistent drift.

- Store historical anomaly events for trend analysis.
- Add geographic mapping for monitoring stations.
- Use domain-specific regulatory thresholds alongside, but separately from, machine-learning anomaly scores.
- Evaluate the system on geographically diverse and independently collected water-quality datasets.
- Introduce model retraining and drift monitoring as new sensor data becomes available.

17. Conclusion

This project demonstrates an end-to-end unsupervised machine-learning pipeline for water-quality anomaly detection. The system processes nine numerical water-quality parameters, handles missing values, standardizes the feature space, trains three complementary anomaly detectors, saves reusable models, generates visual analysis, and provides both CLI and Streamlit interfaces. Isolation Forest serves as the primary detector and flags 164 of the 3,276 observations under the documented 5% contamination setting. The comparison with LOF and One-Class SVM shows that different anomaly-detection methods can identify overlapping but non-identical sets of unusual observations. The resulting system is suitable as an academic demonstration of unsupervised anomaly detection and provides a foundation for future environmental monitoring work.

References

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10. Breunig, M. M., Kriegel, H.-P., Ng, R. T., & Sander, J. LOF: Identifying Density-Based Local Outliers. Proceedings of the 2000 ACM SIGMOD International Conference on Management of Data, 2000.
11. Schölkopf, B., Platt, J. C., Shawe-Taylor, J., Smola, A. J., & Williamson, R. C. Estimating the Support of a High-Dimensional Distribution. Neural Computation, 2001.
12. Pedregosa, F. et al. Scikit-learn: Machine Learning in Python. Journal of Machine Learning Research, 12, 2825–2830, 2011.
13. Kaggle Water Potability Dataset, `water_potability.csv`, as used and documented in the project repository.
14. Streamlit documentation for deployment and application hosting.

Appendix A: Example CLI Flow

```
=====
AI WATER QUALITY ANOMALY DETECTION SYSTEM
=====
CHOOSE AN OPTION:
1. Enter water-quality values manually
2. Use an example NORMAL water sample
3. Use an example ANOMALOUS water sample
```

For a demonstration, option 2 can be used to show a normal observation and option 3 can be used to show an anomalous dataset-based observation. The Streamlit interface provides the same core prediction functionality in a graphical form.